A DISCRETE TIME SIGNAL $s[n]$ IS A SEQUENCE OF NUMBERS DEFINED FOR ALL VALUES OF n.

$s[n]$ IS NOT DEFINED FOR NON-INTEGER VALUES OF n.

$s[3/2]$ IS NOT 0, IT IS UNDEFINED

IT CAN BE REPRESENTED IN MANY WAYS:

Functional

$$s[n] = \begin{cases} 1 & n \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Sequence

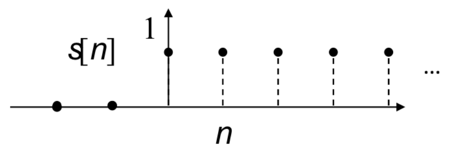
$$s[n] = \{\ldots\ 0\ \ 0\ \ 0\ \ \underset{\uparrow}{1}\ \ 1\ \ 1\ \ 1\ \ldots$$

↑ DENOTES $n=0$

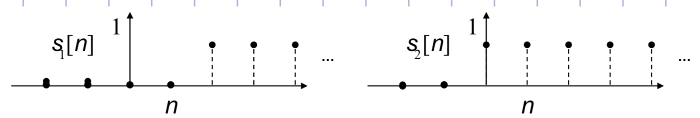
Tabular

n	...	-1	0	1	2	...
$s[n]$	...	0	1	1	1	...

Pictorial



OPERATIONS

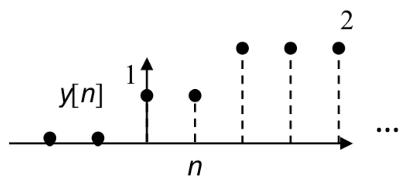


$s_1[n] = [\cdots 0\ \ 0\ 0\ 0\ 1\ 1\ 1\ \cdots]$

$s_2[n] = [\cdots 0\ \ \underset{\uparrow}{0}\ 1\ 1\ 1\ 1\ 1\ \cdots]$

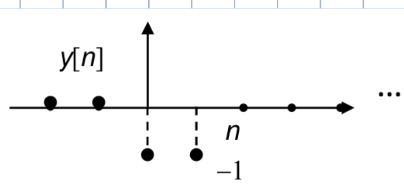
ADDITION

	...	0	0	0 ↓	0	1	1	1	...	+
	...	0	0	1	1	1	1	1	...	
$y[n] = s_1[n] + s_2[n] =$	...	0	0	1	1	2	2	2	...	



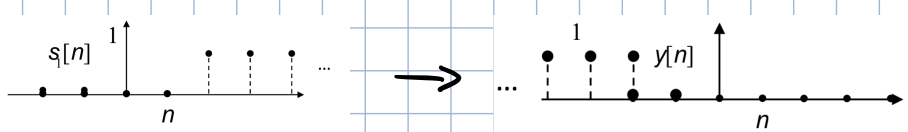
SUBTRACTION

	...	0	0	0 ↓	0	1	1	1	...	−
	...	0	0	1	1	1	1	1	...	
$y[n] = s_1[n] - s_2[n] =$	...	0	0	-1	-1	0	0	0	...	



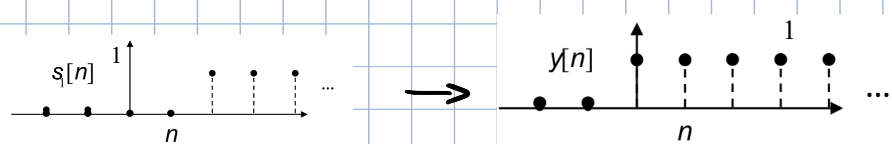
FOLDING / TIME REVERSAL

$$y[n] = s_1[-n]$$



TIME SHIFTING

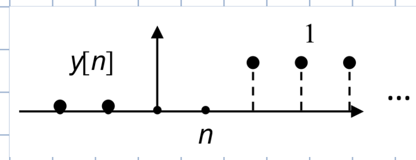
$$y[n] = s_1[n+2]$$



MULTIPLICATION

$Y[n] = S_1[n] \cdot S_2[n]$

···	0	0	$\downarrow$ 0	0	1	1	1	···	+
···	0	0	1	1	1	1	1	···	
···	0	0	0	0	1	1	1	···	



DIVISION

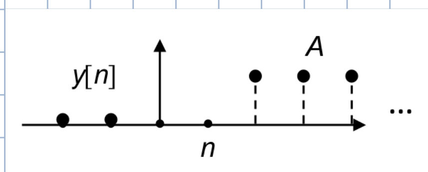
$$Y[n] = \frac{S_1[n]}{S_2[n]}$$

···	0	0	$\downarrow$ 0	0	1	1	1	···	+
···	0	0	1	1	1	1	1	···	
···	/	/	0	0	1	1	1	···	

SCALING

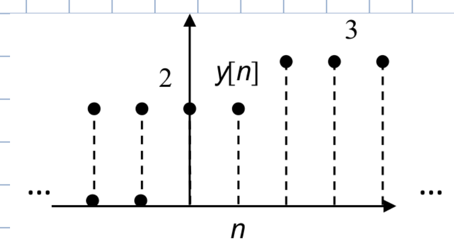
$Y[n] = A\, S_1[n]$

↳ $[\cdots 0\ 0\ \overset{\downarrow}{0}\ 0\ A\ A\ A\ A \cdots]$



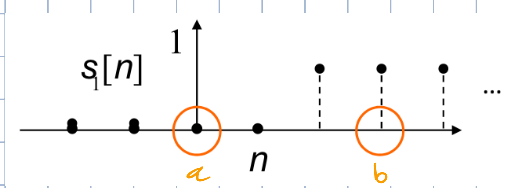
CONSTANT ADDITION

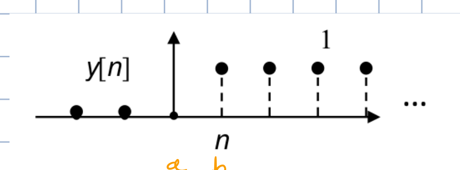
$Y[n] = S_1[n] + A \quad (A=2)$



COMPRESSION

$Y[n] = S_1[An]$





A = 3

PICKS A SAMPLE EVERY 3

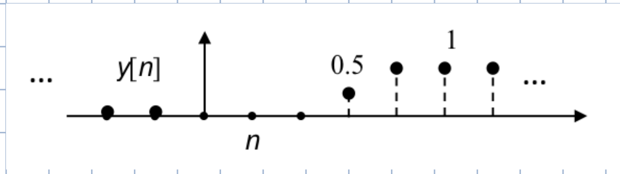
STRETCHING

$Y[n] = S_1\left[\frac{n}{A}\right]$

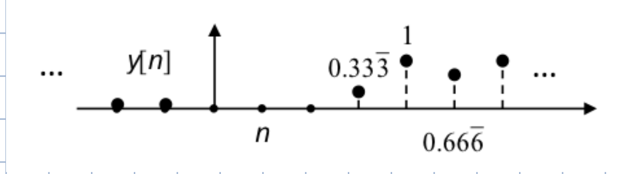
THE STRETCHING OPERATION SAMPLES NON EXISTING VALUES → NEEDS INTERPOLATION

DIFFERENT INTERPOLATION MODELS LEAD TO DIFFERENT RESULTS

$$y[n] = \begin{cases} s\left[\frac{n}{2}\right] & \text{if } n \text{ is even} \\ \dfrac{s\left[\frac{n-1}{2}\right] + s\left[\frac{n+1}{2}\right]}{2} & \text{if } n \text{ is odd} \end{cases} \longrightarrow$$



$$y[n] = \begin{cases} s_i\left[\frac{n}{2}\right] & \text{if } n \text{ is even} \\ \dfrac{s_i\left[\frac{n-3}{2}\right] + s_i\left[\frac{n-1}{2}\right] + s_i\left[\frac{n+1}{2}\right]}{3} & \text{if } n \text{ is odd} \end{cases} \longrightarrow$$

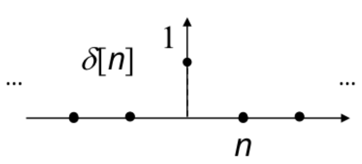


ELEMENTARY SIGNALS

KRONECKER DELTA

SAME AS DIRAC DELTA, BUT IN DISCRETE TIME DOMAIN

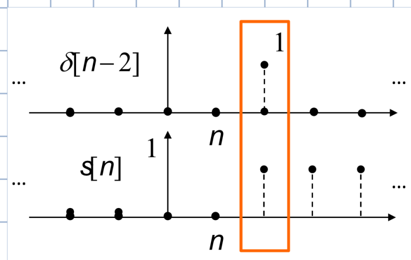
$$\delta[n] = \begin{cases} 1 & n = 0 \\ 0 & \text{otherwise} \end{cases}$$



NOTE: $\delta[n] = \delta[-n]$

! IT CAN BE USED TO SAMPLE A SIGNAL:

$$\sum_{n=-\infty}^{+\infty} s[n] \cdot \delta[n-2] = s[n-2]$$



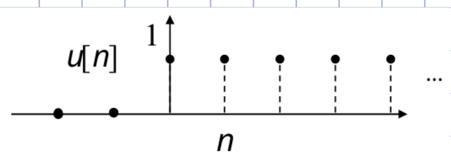
! ANY SIGNAL CAN BE WRITTEN AS AN INFINITE SUM OF PROPERLY SHIFTED AND WEIGHTED DELTAS

$$s[n] = \sum_{k=-\infty}^{+\infty} \delta[n-k]\, s[k]$$

HEAVISIDE / STEP FUNCTION

IT IS AN INFINITE SUM OF DELTAS, STARTING FROM ZERO

$$u[n] = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases}$$



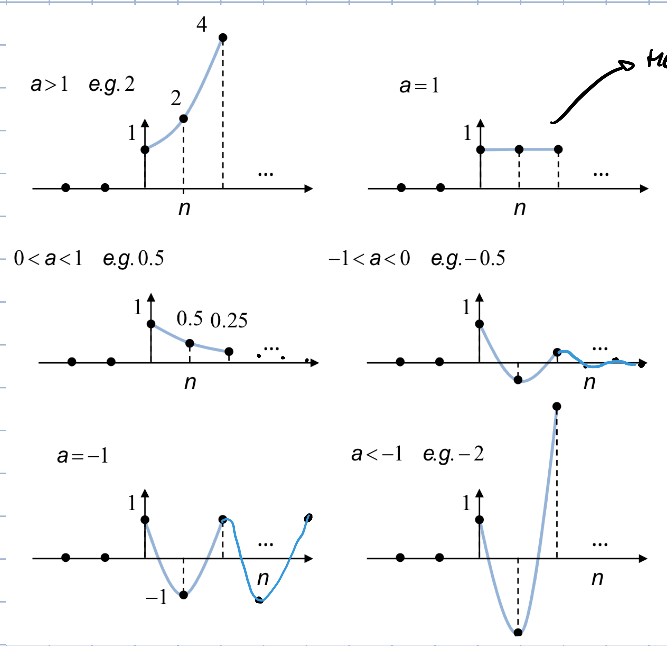
$$u[n] = \sum_{k=0}^{+\infty} \delta[n-k]$$

! IT'S POSSIBLE TO WRITE A DELTA AS A FUNCTION OF TWO HEAVISIDES, THEREFORE EVERY SIGNAL CAN BE REPRESENTED AS A FUNCTION OF SUMS OF HEAVISIDES

$$\delta[n] = u[n] - u[n-1]$$

REAL EXPONENTIAL FUNCTION

$$s[n] = \begin{cases} a^n & n \geq 0 \\ 0 & otherwise \end{cases}$$



COMPLEX EXPONENTIAL FUNCTION

$$s[n] = \begin{cases} a^n = (re^{jw_0})^n = r^n[\cos(w_0 n) + j\sin(w_0 n)] & n \geq 0 \\ 0 & otherwise \end{cases}$$

$|s[n]| = r$

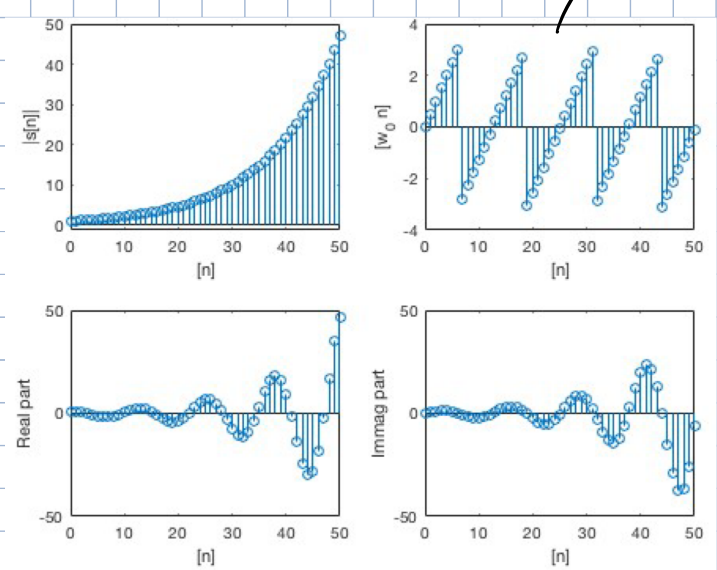
$\angle s[n] = \omega_0 n$ — ω_0 = ANGULAR FREQUENCY

$Re\{s[n]\} = r^n \cos(\omega_0 n)$

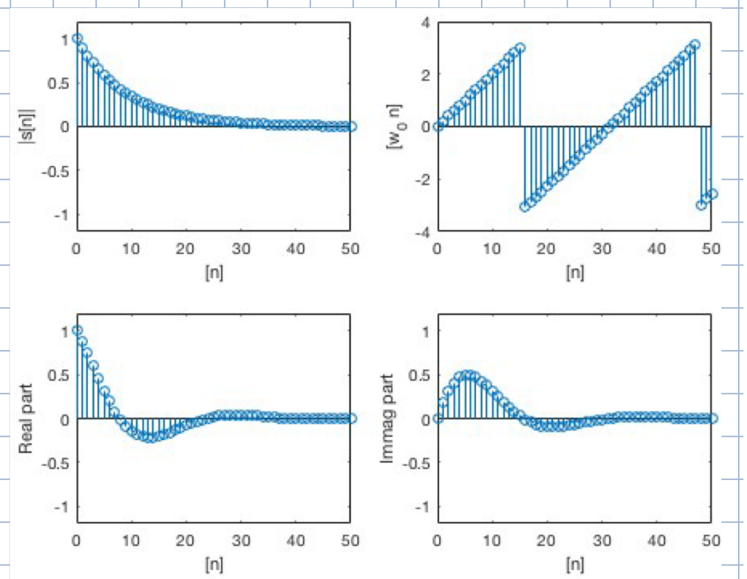
$Im\{s[n]\} = r^n \sin(\omega_0 n)$

ANGULAR !

mod π



$r = 1{,}08 \quad \omega_0 = \frac{1}{2}$



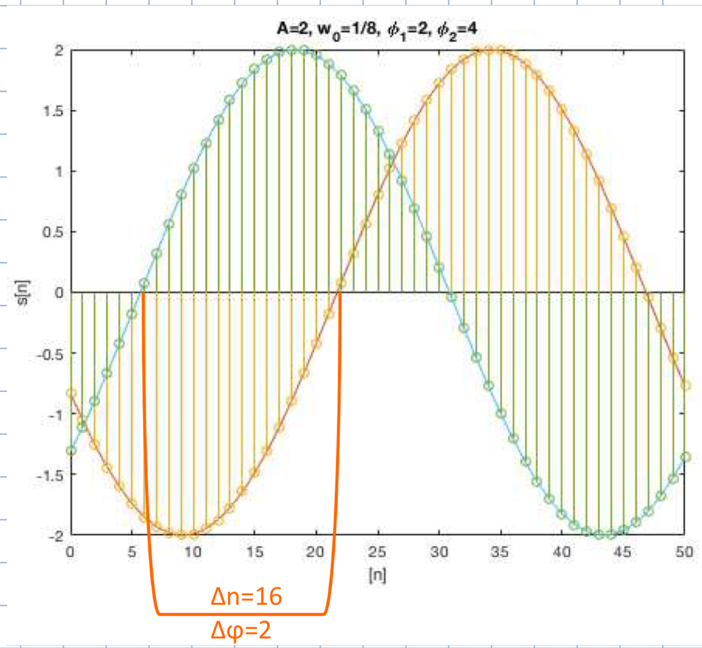
$r = 0{,}9 \quad \omega_0 = \frac{1}{5}$

SINUSOIDAL FUNCTION

$$s[n] = A\cos(w_0 n + \phi)$$

$$\cos(\phi) = \frac{e^{j\phi} + e^{-j\phi}}{2}$$

$$\sin(\phi) = \frac{e^{j\phi} - e^{-j\phi}}{2j}$$



TO COMPUTE PHASE SHIFT:

$A = 2$, $\omega_0 = 1/8$, $\varphi_1 = 2$, $\varphi_2 = 4$

$$\omega_0 n_1 + \varphi_1 = \omega_0 n_2 + \varphi_2$$
$$\downarrow$$
$$\frac{1}{8} n_1 + 2 = \frac{1}{8} n_2 + 4$$
$$\downarrow$$
$$n_1 + 16 = n_2 + 32$$
$$\downarrow$$
$$n_1 - n_2 = 32 - 16 = 16$$

ENZOMMA, RIFROSARE/CAPIRE MEGLIO INTENTO

EVEN FOR VERY HIGH ANGULAR FREQUENCIES, THE SIGNAL REMAINS REPRESENTABLE BY A COS/SIN FUNCTION, AS IT RANGES IN $[-\pi, \pi]$

$$\cos(\omega_0 n) = \cos[(2\pi k + \omega_0) n] = \cos(2\pi k n + \omega_0 n), \quad k \in \mathbb{N}$$

CHECK PERIODICITY

ARE THE FOLLOWING SIGNALS PERIODIC? IF SO, WHAT IS THEIR DISCRETE PERIOD?

AKA HOW MANY SAMPLES DO YOU NEED TO GET A FULL PERIOD OF THE SIGNAL

$$s_1[n] = A\cos(\omega_0 n)\Big|_{\omega_0 = \frac{\pi}{2}}$$

YES, THEY ALL ARE, THEY ARE SINES

$$s_2[n] = A\cos(\omega_0 n)\Big|_{\omega_0 = \frac{\pi}{7}}$$

$$s_3[n] = A\cos(\omega_0 n)\Big|_{\omega_0 = \frac{4}{7}\pi}$$

WE KNOW THAT THE ORIGINAL FUNCTIONS ARE PERIODIC OF 2π, SO TO FIND THE PERIOD WE

NEED TO KNOW WHEN $\omega_0 n = 2\pi$

1) $\omega_0 = \frac{\pi}{2} \longrightarrow \omega_0 n = 2\pi \longrightarrow n = \frac{2\pi}{\omega_0} = 2\cancel{\pi} \cdot \frac{2}{\cancel{\pi}} = 4$

2) $\omega_0 = \frac{\pi}{7} \longrightarrow n = \frac{2\pi}{\omega_0} = 2\cancel{\pi} \cdot \frac{7}{\cancel{\pi}} = 14$

3) $\omega_0 = \frac{4}{7}\pi \longrightarrow n = \frac{2\pi}{\omega_0} = \cancel{2}\cancel{\pi} \frac{7}{\cancel{4}\cancel{\pi}} = \frac{7}{2} \longrightarrow$ 7 SAMPLES AFTER 2 PERIODS OF ORIGINAL SIGNAL
↳ BECOMES PERIODIC OF 7 (4π)

PROPERTIES

BOUNDED SIGNAL

A SIGNAL IS SAID TO BE BOUNDED IF ALL OF ITS VALUES ARE FINITE:

$$|s[n]| \leq \underbrace{B_s}_{\text{BOUNDARY}} < \infty \quad \forall n$$

ABSOLUTE SUMMABILITY

A SIGNAL/SEQUENCE IS ABSOLUTELY SUMMABLE IF THE SUM OF THE ABSOLUTE VALUE ALL SAMPLES IS FINITE:

$$\sum_{n=-\infty}^{+\infty} |s[n]| < \infty$$

ENERGY

$$E_s = \sum_{n=-\infty}^{+\infty} |s[n]|^2$$

IN ELECTRONICS : $P = VI = \frac{V^2}{R} = RI^2$
↳ IF $R = 1 \Rightarrow R = V^2 = I^2$
IN SIGNALS, INSTANTANEOUS POWER : $P_{inst} = x(t)^2$

SQUARE SUMMABILITY / FINITE ENERGY

A SEQUENCE IS SQUARE SUMMABLE, OF FINITE ENERGY, OR AN ENERGY SIGNAL IF ITS ENERGY IS FINITE.

$$E_s < \infty$$

IN OTHER WORDS THE DURATION OF THE SIGNAL IS LIMITED (SINGLE SQUARE PULSE, EXPONENTIAL DECAY, ...).
THE POWER OF AN ENERGY SIGNAL IS ZERO, BECAUSE A FINITE ENERGY IS DIVIDED BY AN INFINITE TIME (LENGTH).

POWER

"POWER IS THE AMOUNT OF ENERGY PER UNIT TIME"
– WIKIPEDIA

$$P_s = \lim_{N \to \infty} \frac{1}{2N+1} \sum_{n=-N}^{N} |s[n]|^2$$

IF THE SIGNAL IS PERIODIC, THE POWER SIMPLIFIES TO

$$P_S = \frac{1}{N}\sum_{n=0}^{N-1} |x[n]|^2$$

POWER SIGNAL

CONTRARY TO AN ENERGY SIGNAL, A POWER SIGNAL IS NOT LIMITED IN TIME, IT EXISTS IN $[-\infty, +\infty]$ (SINEWAVE).
SINCE THE ENERGY OF A POWER SIGNAL IS INFINITE, IT HAS LITTLE MEANING. WE THEREFORE USE POWER (ENERGY PER UNIT TIME), AS IT IS FINITE:

$$0 < P_S < \infty$$

USEFUL MATH FOR POWER SERIES

$$\sum_{n=0}^{\infty}\alpha^n = \frac{1}{1-\alpha} \quad \text{for } |\alpha|<1$$

$$\sum_{n=N}^{\infty}\alpha^n = \frac{\alpha^N}{1-\alpha} \quad \text{for } |\alpha|<1$$

$$\sum_{n=0}^{N-1}\alpha^n = \frac{1-\alpha^N}{1-\alpha} \quad \text{no condition on } \alpha$$

$$\sum_{n=N}^{M-1}\alpha^n = \frac{\alpha^N-\alpha^M}{1-\alpha} \quad \text{no condition on } \alpha$$

ALL DERIVE FROM THIS ONE

EX

$s[n] = u[n]$

1) BOUNDED? $\rightarrow |u[n]| \le 1 < \infty \rightarrow$ YES

2) ABSOLUTELY SUMMABLE? $\rightarrow \sum_{-\infty}^{+\infty} |u[n]| = \sum_{0}^{\infty} 1 = \infty \rightarrow$ NO

3) SQUARE SUMMABLE? $\rightarrow \sum_{0}^{\infty} 1 = \infty \rightarrow$ NO

4) ENERGY? $\rightarrow \sum_{-\infty}^{+\infty} |u[n]|^2 = \sum_{0}^{\infty} 1 = \infty$

5) POWER? $\rightarrow \lim_{N\to\infty} \frac{1}{2N+1}\sum_{-N}^{+N}|u[n]|^2 = \lim_{N\to\infty}\frac{1}{2N+1}\sum_{0}^{N} 1 = \lim_{N\to\infty}\frac{N+1}{2N+1} = \frac{1}{2}$

EX

$s[n] = \alpha^n u[n], \quad 0<\alpha<1$

1) BOUNDED? $\rightarrow |\alpha^n u[n]| \underset{=1 @ n=0}{\le} 1 \rightarrow$ YES

2) ABSOLUTELY SUMMABLE? $\rightarrow \sum_{-\infty}^{+\infty} |\alpha^n u[n]| = \sum_{0}^{\infty} \alpha^n = \frac{1}{1-\alpha} < \infty \iff \alpha \neq 1 \rightarrow$ YES

3) SQUARE SUMMABLE? $\rightarrow \sum_{-\infty}^{+\infty} |\alpha^n u[n]|^2 = \sum_{n=0}^{\infty} \alpha^{2n} \underset{\gamma=\alpha^2}{=} \sum_{n=0}^{\infty}\gamma^n = \frac{1}{1-\gamma} = \frac{1}{1-\alpha^2} \rightarrow$ YES

4) ENERGY? $\rightarrow \frac{1}{1-\alpha^2}$

5) POWER? $\rightarrow \lim_{N\to\infty}\frac{1}{2N+1}\sum_{-N}^{+N}|\alpha^n u[n]|^2 = \lim_{N\to\infty}\frac{1}{2N+1}\sum_{0}^{N}\alpha^{2n} = \lim_{N\to\infty}\frac{1}{2N+1}\cdot\frac{1}{1-\alpha^2} = 0$

EX

$s[n] = \alpha^n u[n]$, $\alpha = 1 \rightarrow s[n] = 1^n u[n] = u[n]$

1) BOUNDED ? $\rightarrow |u[n]| \leq 1 \rightarrow$ YES

2) ABSOLUTELY SUMMABLE ? $\rightarrow \sum_{-\infty}^{+\infty} |u[n]| = \infty \rightarrow$ NO

3) SQUARE SUMMABLE ? $\rightarrow \sum_{-\infty}^{+\infty} |u[n]|^2 = \infty \rightarrow$ NO   IT WOULD BE IF $|\alpha| < 1$

4) ENERGY ? $\rightarrow \infty$

5) POWER ? $\rightarrow \lim_{N\to\infty} \frac{1}{2N+1} \sum_{-N}^{N} |u[n]|^2 = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{0}^{N} 1 = \lim_{N\to\infty} \frac{N+1}{2N+1} = \frac{1}{2}$

LENGTH

A DISCRETE TIME SIGNAL CAN BE A SEQUENCE OF FINITE OR INFINITE LENGTH.

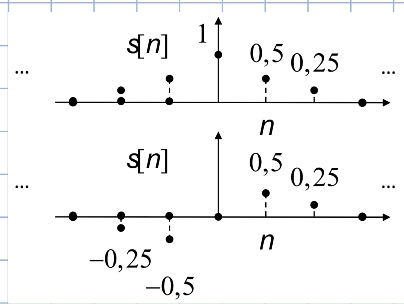
AN INFINITE LENGTH SEQUENCE CAN BE:

- RIGHT SIDED: $s[n] = 0 \ \forall n < N$, $N \in \mathbb{Z}$ (E.G. $u[n]$)
- LEFT SIDED: $s[n] = 0 \ \forall n > N$, $N \in \mathbb{Z}$ (E.G. $u[-n]$)
- BOTH SIDED: NEITHER LEFT NOR RIGHT SIDED (E.G. $u[n] + u[-n]$)

SYMMETRY

A DISCRETE TIME SIGNAL CAN BE ODD OR EVEN

- EVEN : $x[n] = x[-n]$
- ODD : $x[n] = -x[-n]$



EVERY SIGNAL CAN BE WRITTEN AS THE SUM OF ITS EVEN AND ODD PARTS

$$s[n] = s_{EVEN}[n] + s_{ODD}[n]$$

$$s_{EVEN}[n] = \frac{s[n] + s[-n]}{2} \qquad s_{ODD}[n] = \frac{s[n] - s[-n]}{2}$$

NORMALIZATION FACTOR

ARTIFICIAL EVEN REPRESENTATION OF THE SIGNAL. COPY RIGHT SIDE ON LEFT SIDE TO FORCE EVEN SYMMETRY

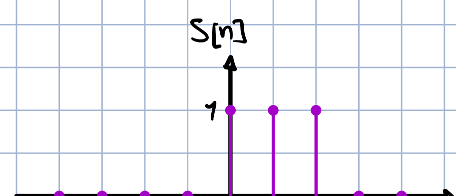
SAME BUT FORCED ODD REPRESENTATION

EX

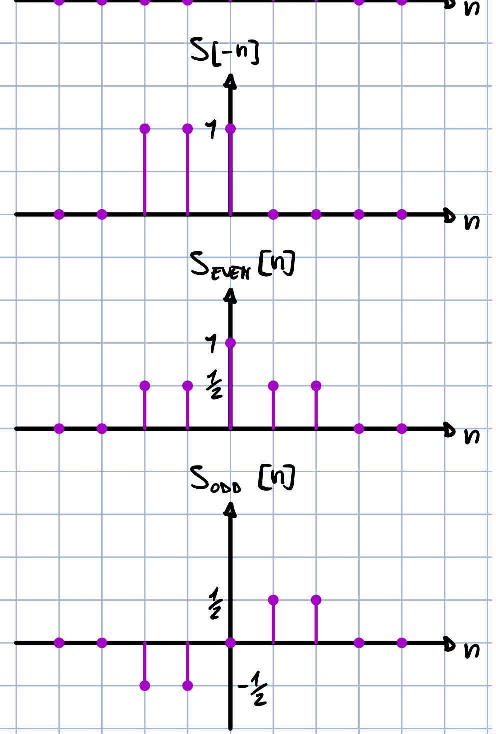
$s[n] = \delta[n] + \delta[n-1] + \delta[n-2]$

$\delta[n-2] \rightarrow \delta[-n-2] = \delta[-(n+2)] = \delta[n+2]$

$\delta[n] = \delta[-n]$



$$S[-n] = \delta[n+2] + \delta[n+1] + \delta[n]$$



$$S_{EVEN}[n] = \frac{\delta[n]}{2} + \frac{\delta[n-1]}{2} + \frac{\delta[n-2]}{2} + \frac{\delta[n+2]}{2} + \frac{\delta[n+1]}{2} + \frac{\delta[n]}{2}$$

$$= \delta[n] + \frac{\delta[n-1]}{2} + \frac{\delta[n-2]}{2} + \frac{\delta[n+2]}{2} + \frac{\delta[n+1]}{2}$$

$$S_{ODD}[n] = \cancel{\frac{\delta[n]}{2}} + \frac{\delta[n-1]}{2} + \frac{\delta[n-2]}{2} - \frac{\delta[n+2]}{2} - \frac{\delta[n+1]}{2} - \cancel{\frac{\delta[n]}{2}}$$

$$= \frac{\delta[n-1]}{2} + \frac{\delta[n-2]}{2} - \frac{\delta[n+2]}{2} - \frac{\delta[n+1]}{2}$$

$$S_{EVEN}[n] + S_{ODD}[n] = \delta[n] + \frac{\delta[n-1]}{2} + \frac{\delta[n-2]}{2} + \cancel{\frac{\delta[n+2]}{2}} + \cancel{\frac{\delta[n+1]}{2}} +$$

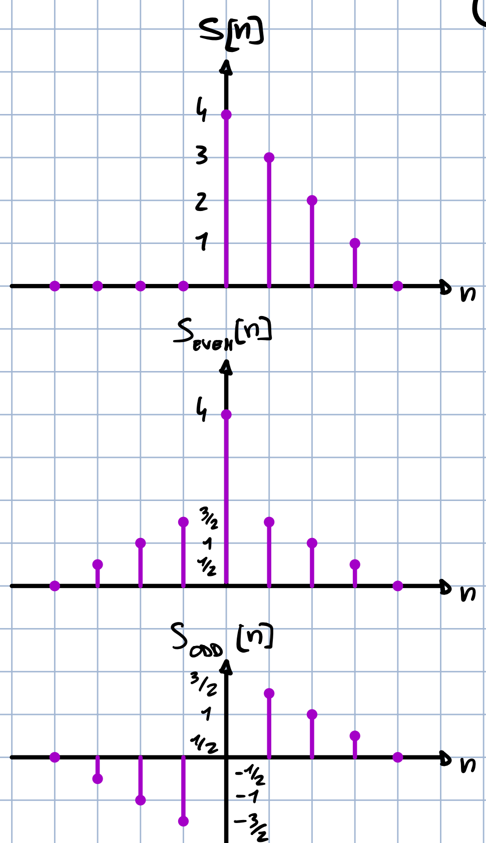
$$+ \frac{\delta[n-1]}{2} + \frac{\delta[n-2]}{2} - \cancel{\frac{\delta[n+2]}{2}} - \cancel{\frac{\delta[n+1]}{2}}$$

$$= \delta[n] + \delta[n-1] + \delta[n-2] = S[n] \quad \#$$

EX

FIND THE ODD AND EVEN PARTS OF THE FOLLOWING SIGNAL:

$$S[n] = \begin{cases} 4-n & 0 \le n \le 4 \\ 0 & \text{OTHERWISE} \end{cases}$$



$$S[n] = 4\delta[n] + 3\delta[n-1] + 2\delta[n-2] + \delta[n-3]$$

$$S[-n] = \delta[n+3] + 2\delta[n+2] + 3\delta[n+1] + 4\delta[n]$$

$$S_{EVEN}[n] = \frac{S[n] + S[-n]}{2} = 4\delta[n] + \frac{3}{2}\delta[n-1] + \delta[n-2] + \frac{1}{2}\delta[n-3] +$$

$$+ \frac{1}{2}\delta[n+3] + \delta[n+2] + \frac{3}{2}[n+1]$$

$$S_{ODD}[n] = \frac{S[n] - S[-n]}{2} = \frac{3}{2}\delta[n-1] + \delta[n-2] + \frac{1}{2}\delta[n-3] - \frac{1}{2}\delta[n+3]$$

$$- \delta[n+2] - \frac{3}{2}[n+1]$$